

GSC26

Genomic Standards in Indigenous and AI contexts

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Agenda

Day 1: Wednesday August 12 - Genomic Standards in Indigenous contexts

Themes: Oceania genomics & Indigenous metadata; Access and benefit sharing and biocultural stewardship; Regulations

8:30	Check In - Mokoia Room
8:45-9:30	Formal Welcome (Mihi whakatau) and Introduction - Ngāti Whakaue (Māori hosts) GSC Welcome: Lynn Schriml , GSC President (University of Maryland School of Medicine) New Zealand Welcome: Libby Liggins (University of Auckland, Aotearoa Genomic Data Repository) Claire Rye (University of Auckland, Aotearoa Genomic Data Repository) Mik Black (Genomics Aotearoa) Alana Alexander (University of Otago)
9:30-10:00	Morning Break
Session 1:	<u>Data Standards in Indigenous contexts</u>
10:00-10:30	Keynote: Amanda Black (Lincoln University) Indigenous data sovereignty, Māori data sovereignty, and applications to genomics
10:30-10:50	Cas Sedran-Price (GIDA & The University of Sydney) The CARE Principles for Indigenous Data Governance
10:50-11:10	Leke Hutchins (University of California, Berkeley) Applying a CARE Informed Data Lifecycle Approach to DNA Metabarcoding and eDNA
Session 2:	<u>Operationalizing CARE/FAIR in genomics</u>
11:10-11:30	Kalinda Griffiths (Poche SA+NT, Flinders University) Developing an Indigenous Genomic Data Stewardship Methodology: Lessons from Australia's CONNECT Program
11:30-11:50	Platinum Sponsor Kris Locken (Zymo Research) Platinum Sponsor From Metadata to Measurement: A Framework for Workflow Standards in Microbiome Science
12:00-1:00	Lunch
1:00-1:20	Tracey Godfery (Genomics Aotearoa, University of Otago) CARE in practice: community aspirations realised & empowered within an Indigenous Genomics Platform
1:20-1:40	Huti Watson (Ngāti Porou Oranga) Claire Rye (University of Auckland, Aotearoa Genomic Data Repository) Rakeiora: A pathfinder for genomic medicine in Aotearoa/New Zealand
1:40-2:00	Katharina Ruckstuhl (University of Otago) Local Contexts Notices and Labels: reasserting cultural authority through digital infrastructure
2:00-2:20	Neil Davies (UC Berkeley) Moorea Biocode and Ethnocode Projects
2:20-2:40	Tui Kaumoana (University of Auckland) and Michael Steedman (University of Auckland)
3:00-3:30	Afternoon Break
3:30-4:30	Panel Discussion: Alana Alexander (Chair): Data Standards in Indigenous Contexts
4:30-5:00	Group Photo
5:30-7:00	Welcome Reception & Posters

Day 2: Thursday August 13 - Genomic Standards

Themes: GSC Standards; Single-cell, eDNA, ancient DNA Standards; Standards Across Genomic Repositories & DNA reference databases: Partnership models & LISTEN principles

8:30	Check In
8:45	Welcome
Session 3:	<u>GSC Standards & Projects</u>
9:00-9:20	Montana Smith (NMDC) Updates to MIxS and v7.0
9:20-9:40	Tina Warinner (Harvard University) Ancient DNA (aDNA) MInAS update MINAS and open access resources for ancient metagenomics research and education
9:40-10:00	Luke Thompson (NOAA) NOAA eDNA data workflow in the context of NOAA Ocean Exploration
10:00-10:30	Morning Break
10:30-11:00	Dawn Field Award - Miwa Takahashi (CSIRO) FAIR eDNA checklists FAIR eDNA metadata checklists
11:00-11:10	Katie Pitz (virtual) (MBARI) BeBOP protocols: Better Biomolecular Ocean Practices through standard formatting, field-specific metadata, and archiving with version control
11:10-11:30	Tanja Woyke (DOE, Joint Genome Institute) Microeukaryote single-cell genomes: time for standards
11:30-11:40	Pier Luigi Buttigieg (virtual) (Alfred Wegener Institute) Fusing MIxS into cross-domain knowledge graphs: Observations and implications for the future of the standard.
11:45-12:45	Lunch
Session 4:	<u>Omic Observing and Reference Databases</u>
1:00-1:20	Andrew Bissett (CSIRO) Australian Microbiome: Standards, checklists and incentivising open microbiome data
1:20-1:40	Tobias Frøslev (GBIF) Global Biodiversity Information Facility
1:40-2:00	John Deck (UC Berkeley) Genomic Observatories Metadatabase (GeOMe)
2:00-2:20	Chris Meyer (Smithsonian's National Museum of Natural History) A Roadmap toward developing trusted reference databases
2:30-3:00	Afternoon Break
Session 5:	<u>Sequence Repositories</u>
3:00-3:20	Claire Rye (University of Auckland, Aotearoa Genomic Data Repository) Mik Black (Genomics Aotearoa) Furthering Kaitiakitanga in the Aotearoa Genomic Data Repository
3:20-3:40	Masanori Arita (National Institute of Genetics, Japan) DDBJ and DBCLS: a big boost toward a national database center
3:40-4:00	Guy Cochrane (EMBL-EBI, Global Biodata Coalition) Standards: Enabling sustainability in global biodata infrastructure
4:00-4:10	Peter Woollard (virtual) (European Bioinformatics Institute EMBL-EBI) ENA: The future eDNA digital landscape and implications for standards bodies and repositories
4:10-4:30	Seungwoo Hwang (Korea Bioinformation Center) Introduction to Korea BioData Station
4:30-4:50	Yuriy Skripchenko (National Center for Biotechnology Information (NCBI), SRA) The INSDC Specifications: Enabling Global Participation and Interoperable Sequence Data
5:00-6:00	Panel Discussion: Mik Black (Chair) Sequence Repositories: How to work together within an open system, maintaining interoperability between open and controlled access data repositories
6:00 -7:00	Evening Networking: Thursday night market

Day 3: Friday August 14 - Ethical, Legal, and Social Context

Themes: Ethical, Legal, and Social metadata; Standards for Bioconvergence: Genomics and AI

8:30 Check In
8:45 Welcome

Session 6: Building trustworthy omic data ecosystems

- 9:00-9:20** **Ann Mc Cartney** (University of California, Irvine)
A Convention on Biological Diversity commissioned study on tools and models, such as databases, for making the digital sequence information on genetic resources publicly available and accessible in a transparent and accountable manner
- 9:20-9:40** **Erena Mikaere** (Te Kāhui Raraunga)
Te Pā Tūwatawata: Distributed Storage Infrastructure for Māori Data Sovereignty. Tikanga and kawa guiding the storage treatment of data
- 9:40-10:00** **Joseph Yracheta** (Native Biodata Consortium)
Indigenous Peoples DATA and Controlling Repositories: Advancing #IDSov and Thwarting AI Genocide
- 10:00-10:20** **Maheata White Davies** (Université de la Polynésie Française)
Te Ora Fenua - Te Mauri Ora: ethical, legal, social aspects of place-based data trusts
- 10:20-11:00** Morning Break
- Session 7: Omics in AI and Digital Twins**
- 11:00-11:20** **Manu Caddie** (Te Kotahi Research Institute)
Beyond the Repository: Federated Indigenous Governance Infrastructure for Biodiversity Data
- 11:20-11:40** **Katherine Tattersall** (CSIRO)
Building Digital Systems for Marine Genetic Samples and eDNA Biodiversity Data: Trusted Architectures, AI-Enabled Publishing, and International Interoperability
- 11:40-11:50** **Christina Pavloudi** (*virtual*) (EMBRC - European Marine Biological Resource Centre)
Building the genomics data flow to the European Digital Twin Ocean [10 min recorded]
- 12:00-12:15** Handoff GSC27 & Meeting Closing
- 12:15-1:30** Lunch
- 1:30-3:30** **Workshop: Indigenous metadata standards**: fields to capture provenance, associated knowledge, rights, and permissions
- 3:30-4:00** Afternoon Break
- 4:00-5:00** **Workshop: Indigenous metadata standards** resumes

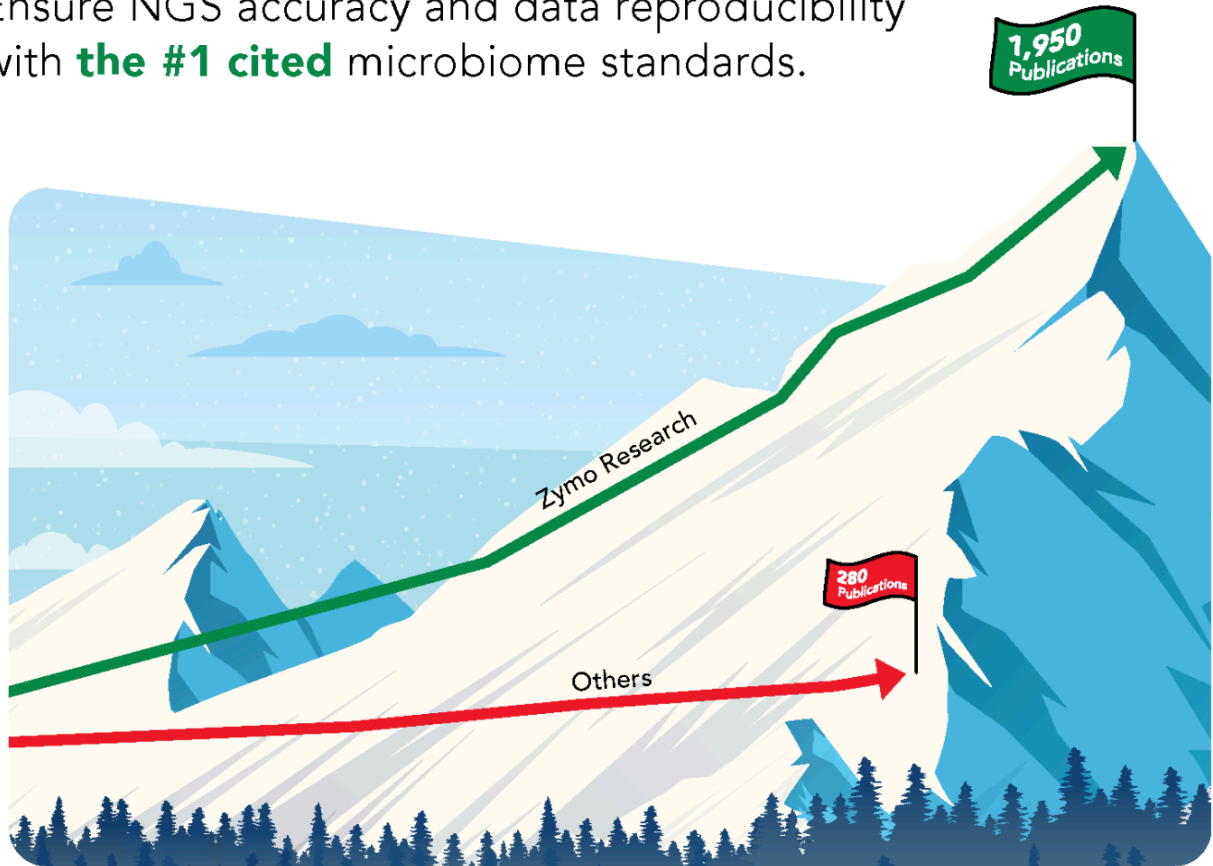
Posters

1. **Jiwon An:** Introduction to the Korea BioData Station (K-BDS): A Biological Data Repository for Standardized Data Sharing and Reuse
2. **Maria Chuvochina:** Genomic standards for eukaryotes: Time for an update?
3. **Nik Palevich:** Metagenomic insights into rumens adapted to a climate-smart forage
4. **Yue Yin:** Uncovering distinct ecological processes and key drivers shaping indoor microbiomes in detached houses and apartments
5. **Jingjing Zhang:** Enabling Māori Data Governance and access to taonga data
6. **Byungk Lee:** Introduction to the Korea BioData Station (K-BDS): A Biological Data Repository for Standardized Data Sharing and Reuse
7. **Woori Chae:** Bridging Metabolomics into Standardized BioData Infrastructure: KMAP and an Automated Multi-vendor Quality-Control Pipeline
8. **Natalie Forsdick:** Journeying towards best practice: Tips and tricks for improved data management for biodiversity genomics researchers
9. **Jorja Hutton:** Developing a Functional Indigenous Stewardship Model for Genomics Data in Australia
10. **Selina Patel:** The recovery of human environmental DNA



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